NanoGT Match Report: Hemophilia B

Disease: Hemophilia B (ORPHA:306)
Primary gene: F9
Gene CDS: 1383 bp
Inheritance: X-linked recessive
Target tissues: liver

Top 5 GT Precedent Matches

Rank	Program	Vector	Score	Confidence	Approval
1	Hemgenix	AAV5	9.9/10	[GREEN] High	approved
2	Roctavian	AAV5	9.9/10	[GREEN] High	approved
3	SPK-8011	AAVrh10	9.5/10	[GREEN] High	phase3
4	DTX201	AAV8	9.1/10	[GREEN] High	phase2
5	ST-920	AAV2/6	8.5/10	[GREEN] High	phase1/2

Match #1: Hemgenix

Precedent disease: Hemophilia B
Vector: AAV5
Tissue target: liver
Composite score: 9.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?

Dimension	Score	Max	What it measures
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**9.89**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #2: Roctavian

Precedent disease: Hemophilia A
Vector: AAV5
Tissue target: liver
Composite score: 9.9 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue

Dimension	Score	Max	What it measures
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	1.00	1.0	Regulatory approval / trial stage
Immunogenicity	2.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**9.89**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Approval status: approved
- Vector immunogenicity (AAV5): low (~9%) — most patients eligible; minimal screening burden
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #3: SPK-8011

Precedent disease: Hemophilia A
Vector: AAVrh10

Tissue target: liver
Composite score: 9.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.80	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**9.50**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation
- Approval status: phase3
- Vector immunogenicity (AAVrh10): moderate (~10%) — significant proportion may require pre-screening or exclusion
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)

- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #4: DTX201

Precedent disease: Hemophilia A
Vector: AAV8
Tissue target: liver
Composite score: 9.1 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	2.00	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.60	1.0	Regulatory approval / trial stage
Immunogenicity	1.00	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue
TOTAL (normalised)	**9.11**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Pathway match: coagulation

- Approval status: phase2
- Vector immunogenicity (AAV8): high (~30%) — substantial patient exclusion expected; immunodepletion protocols may be needed
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs

Match #5: ST-920

Precedent disease: Fabry disease
Vector: AAV2/6
Tissue target: liver
Composite score: 8.5 / 10

Score Breakdown

Dimension	Score	Max	What it measures
Packaging fit	2.00	2.0	Gene CDS size vs vector cargo capacity
Tissue tropism	2.00	2.0	Vector naturally reaches disease target tissue
Protein class	2.00	2.0	Same secreted/lysosomal/membrane/intracellular class
Pathway similarity	0.50	2.0	Same or related biological pathway
Inheritance compatibility	1.00	1.0	AR/XL loss-of-function pattern match
Approval precedent	0.50	1.0	Regulatory approval / trial stage
Immunogenicity	1.50	2.0	Pre-existing NAb seroprevalence for this vector
Therapeutic window	2.00	2.0	Can GT be given before irreversible damage?
Cross-correction	1.00	1.0	Can transduced cells rescue untransduced neighbours?
Immune privilege	0.80	1.0	Immunological protection of target tissue
Promoter availability	1.00	1.0	Validated tissue-specific promoters exist
Route of administration	1.00	1.0	Established delivery route to target tissue

Dimension	Score	Max	What it measures
TOTAL (normalised)	**8.50**	**10.0**	Raw sum / 18 × 10

Rationale

- Gene CDS 1383bp / cargo 4700bp (29% utilized)
- Vector tropism plus precedent target match: liver
- Both secreted proteins — systemic delivery viable
- Inheritance match (X-linked recessive <-> XL)
- Different pathway (coagulation vs lysosomal_storage)
- Approval status: phase1/2
- Vector immunogenicity (AAV2/6): moderate (~17%) — significant proportion may require pre-screening or exclusion
- Wide therapeutic window — adult or chronic onset; GT can be administered at multiple timepoints; irreversible damage has not yet occurred at typical diagnosis age
- Secreted protein — systemic cross-correction via bloodstream; all cells can benefit from a minority of transduced cells
- Immune privilege: moderate-high privilege — tolerogenic microenvironment (Kupffer cells, IL-10, PD-L1)
- Promoter availability: ApoE/hAAT, TBG, transthyretin, albumin — extensively validated; used in Hemgenix, Roctavian, DTX301
- Route of administration: IV systemic — established, minimally invasive; used in all hepatic GT programs